

Topic 8: How AI Models are Trained

8.1 Pre-Training on Large Corpora

- Pre-training: the model learns by predicting the next token on an enormous dataset of text from the internet, books, and code
- Data sources: Common Crawl (web pages), The Pile, Wikipedia, GitHub code, books, scientific papers
- Self-supervised: no human labelling needed -- the 'label' is the next token, which already exists in the text
- Scale: GPT-3 was trained on ~300 billion tokens; modern frontier models use trillions
- Pre-training gives the model broad world knowledge and language understanding -- it becomes a 'base model'

8.2 Fine-Tuning and RLHF

- Fine-tuning: continue training the base model on a smaller, curated dataset to specialise its behaviour
- Supervised fine-tuning (SFT): show the model high-quality example conversations written by human contractors
- RLHF (Reinforcement Learning from Human Feedback): human raters compare model outputs, training a reward model that scores responses
- The language model is then optimised using PPO (Proximal Policy Optimisation) to maximise the reward model's scores
- RLHF is responsible for making base models into useful, aligned assistants that follow instructions safely
- InstructGPT paper introducing RLHF: <https://arxiv.org/abs/2203.02155>

8.3 Compute and Infrastructure

- Training frontier models requires specialised hardware: NVIDIA A100 or H100 GPUs, or Google TPUs
- Training GPT-3 was estimated to require thousands of A100 GPUs running for weeks
- Data centres for AI training require enormous amounts of electricity and specialised cooling
- Inference (serving the model to users) is much cheaper than training -- can use smaller, optimised hardware
- Quantisation: reducing model weight precision (e.g., 16-bit to 4-bit) reduces memory and speeds up inference
- Energy and water consumption of AI training is a growing environmental concern; major labs publish sustainability reports

8.4 Parameter-Efficient Adaptation

- Full fine-tuning: update all model parameters on a new dataset -- very expensive for large models
- LoRA (Low-Rank Adaptation): add small trainable weight matrices to frozen model layers -- 10--100x fewer trainable parameters
- QLoRA: quantise the base model to 4-bit precision, then apply LoRA -- enables fine-tuning on consumer GPUs
- RAG (Retrieval-Augmented Generation): don't train at all -- retrieve relevant documents at query time and include them in the prompt
- For most applications, RAG or prompt engineering is more practical than fine-tuning

```
# LoRA fine-tuning with Hugging Face PEFT library:
from peft import LoraConfig, get_peft_model
from transformers import AutoModelForCausalLM

base_model = AutoModelForCausalLM.from_pretrained('meta-llama/Llama-3-8b')
lora_config = LoraConfig(r=16, lora_alpha=32, target_modules=['q_proj', 'v_proj'])
model = get_peft_model(base_model, lora_config)
model.print_trainable_parameters() # ~0.1% of total parameters
```